

1) A1 Encrypt "DEFEND THE WALL" with Key = 3.

D → G, E → H, F → I, E → H, N → Q, D → G.

Ciphertext: GHIHQG WKH ZDOO

2) Decrypt "KHOBK ZRVOG" with Key = 3

K → I, H → E, O → L, O → LR → O

Plaintext: HELLO WORLD.

3) "SECRET" → "VHFUHW".

S → V, E → H, C → F

Decrypt: "WKLV LVD WHVW"

W → T, K → H, L → I, V → S

Plaintext: THIS IS A TEST

2) A1 Encrypt "FIRE"

F → Y, I → O, R → K, E → T

Ciphertext: YOKT

2) Decrypt "UJHHD QEB NRFZW YOLTK"

Using frequency Analysis.

Plaintext: ATTACK THE SOUTH GATE.

3) A1 Encrypt "ATTACKATPALACE" using Keyword "LEMON".

Plaintext    A T T A C K A T P A L A C E

Key        L E M O N L E M O N L E M O

Ciphertext: L X F O P V E F R N H R M H

② Decrypt "LXF-OPVEFRNH R" using "LEMON".

Plaintext: ATTACKATDAWN.

③ Decrypt "QNXEPKMAEGTU" using key "KEY".

Original Message: GDTALFCWACPK.

4)A)

Key Matrix:

①

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

1. Encrypt "HI"

H=7, I=8

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 45 \\ 84 \end{bmatrix}$$

$$45 \text{ Mod } 26 = 19 = T$$

$$84 \text{ Mod } 26 = 2 = C$$

Ciphertext: TC

② Verify "PO" → "QC"

P=15, O=14

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 15 \\ 14 \end{bmatrix} = \begin{bmatrix} 87 \\ 100 \end{bmatrix}$$

$$87 \text{ Mod } 26 = 9 = J$$

$$100 \text{ Mod } 26 = 22 = W$$

Result JW (not QC).

5)A)

① Encrypt "MEETMEATERLUNCH" using key word "ZEBRA".

Ciphertext: LFKMEZJXISNMRY.

② Decrypt "EANNTEFCUHTREL".

Using Key word "BOARD".

Plaintext: ENCRYPTONMETHOD.

6) A) Plaintext: HELLO

① Key : XMCKL

using . A=0, B=1, ..., Z=25.

| Letter | P  | Key   | $(P+K) \bmod 26$ | Cipher |
|--------|----|-------|------------------|--------|
| H      | 7  | X(23) | 4                | E      |
| E      | 4  | M(12) | 16               | Q      |
| L      | 11 | C(2)  | 13               | N      |
| L      | 11 | K(10) | 21               | V      |
| O      | 14 | L(11) | 25               | Z      |

Cipher text : EQNVZ

②

→ Key is random.

→ Key length = message length.

→ Key is used only once.

→ Makes guessing the plaintext impossible

7) A) (A) Encrypt "WEAREDISCOVERED" (Key word: ZEBRA).

Matrix :

|   |   |   |   |   |
|---|---|---|---|---|
| Z | E | B | R | A |
| W | E | A | R | E |
| D | I | S | C | O |
| V | E | R | E | D |

Column order: A, B, E, R, Z

Cipher text:

EODASREIERCEWDV.

⑧ INFORMATION SECURITY (Keyword: HACK)

③ → Rows Needed = 5

→ Padding = X

Ciphertext: NMOCIFANUYIRLELOTSRX

③ Cipher text:

EVLNEACOTKESEA@ROFOTDEBCOXWN.

(Keyword: TRIA)

Plain text :

WEARDISCOVEREDFLEEATONCE.

8) A)

Most Common LETTERS:

|         |         |
|---------|---------|
| • O = 8 | • U = 5 |
| • Z = 8 | • S = 5 |
| • P = 5 |         |

Probable Mapping:

|         |         |
|---------|---------|
| • O → E | • P → O |
| • Z → T | • S → I |
| • U → A |         |

Partial Decryption:

ATEIEVAEHXMEOVO...

Best Guess: Message is likely English, but more Substitutions are needed for full Decryption.

9) A) Ciphertext : BTJW QJW YMTW NX FXNRQJWX

Known plaintext: THIS IS THE

Mapping:

|         |         |         |
|---------|---------|---------|
| • B → T | • W → I | • M → E |
| • J → H | • Q → S |         |

Decrypted Text: THIS IS THE AN EXAMPLE

10) A)

Ciphertext:

ZI CVTWQNRZGNTWAVZHCQYGLMGJ.

Key: KEY

Plain text:

CRYPTOGRAPHY IS FUN AND LEARNING.